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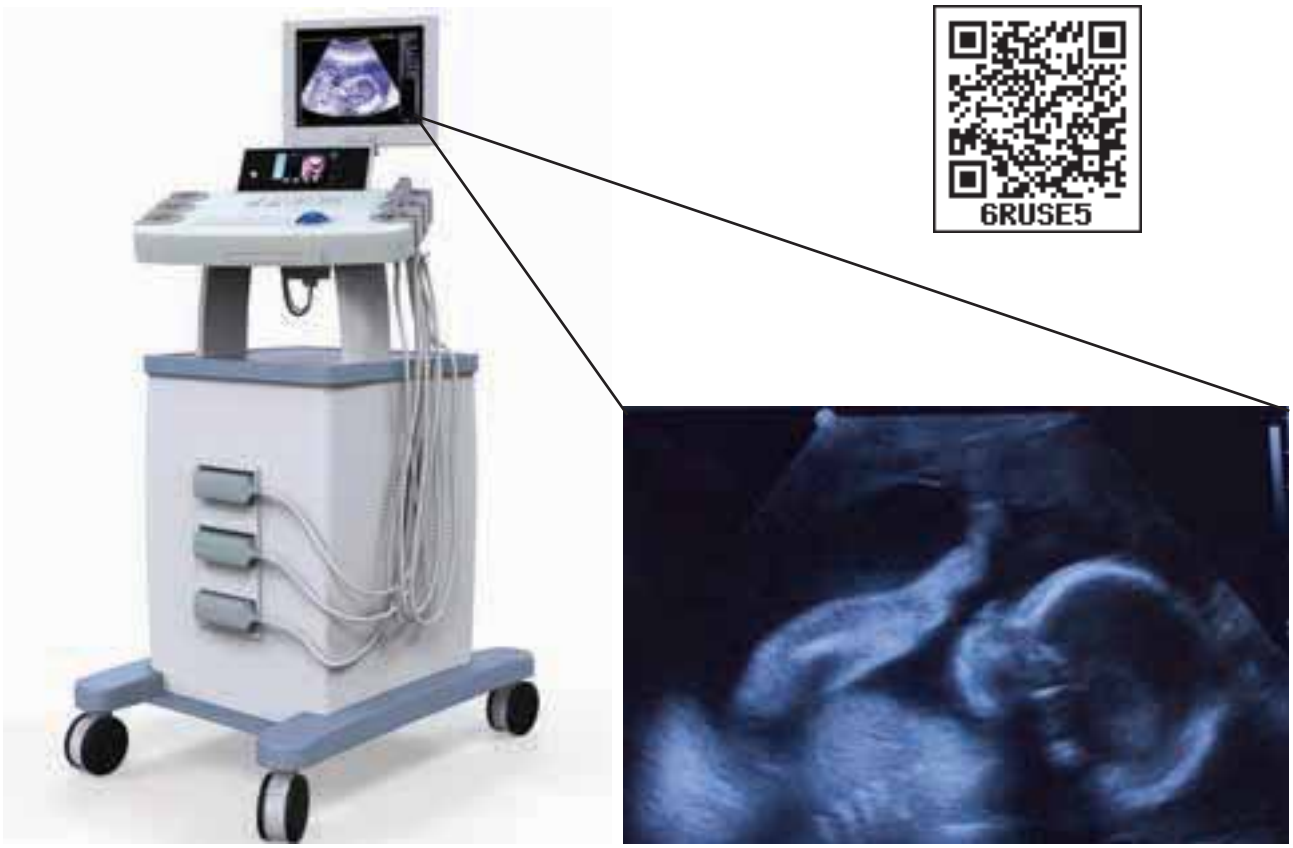


**Figure 12.7.** Ultrasonography

4. Mammography is a special type of X-ray imaging to create detail images of the breast tissues
5. Fluoroscopy for real time images
6. X-rays are used in radiation therapy to shrink cancerous tumours

## Ultrasound imaging

Ultrasound literally means sound beyond the range of human hearing. Ultrasound waves are produced by a physical phenomenon known as Piezo-electric effect. When an electric potential is applied to certain crystals for example: Lead zirconate, they become excited, vibrate and produce ultrasound. When the ultrasound waves are introduced through homogenous tissue, they pass unimpeded until they meet another tissue or organ. A part or whole of the ultrasound wave is reflected and received back by the same crystal and is converted into an electrical signal. This signal denoting reflecting interface is shown on the oscilloscope screen as a deflection from the base line (Figure12.7).

## Clinical significance

1. Ultrasound waves are used to image the foetus at different stages of pregnancy to study the progress of the developing foetus.
2. They are used to hear foetal heart sound, blood flow, etc.
3. Used in echocardiography to diagnose the damages in heart.
4. Used for diagnosis of tumours, gall stones, kidney stones, obstructions in the genital tracts.